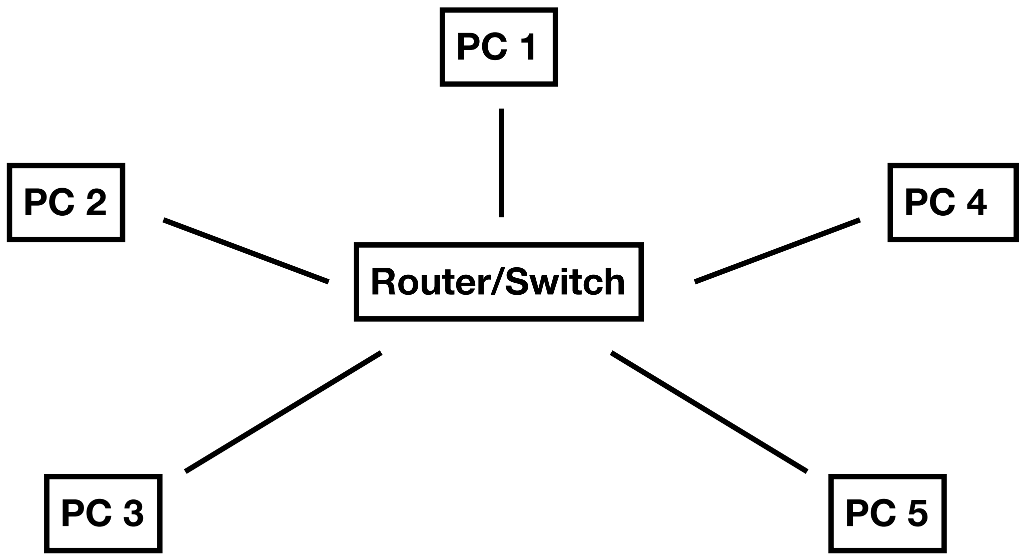
# Module 1: Network + Network Fundamentals and building networks

## Network Topologies

### Task 1:

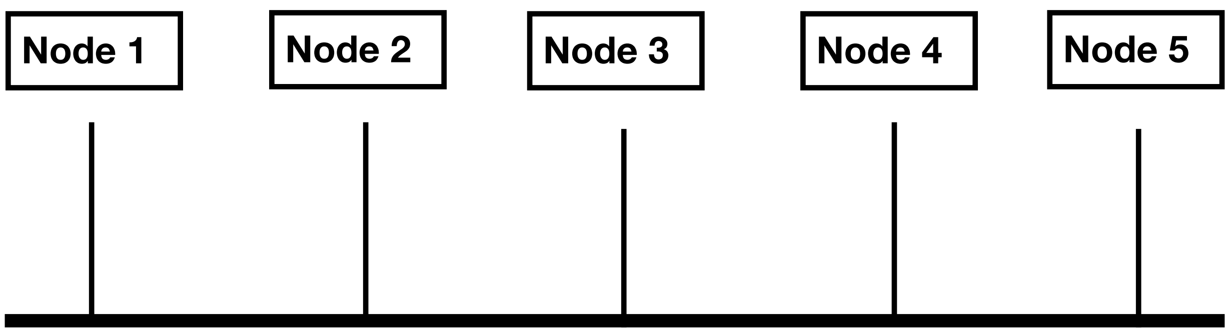
**1.**




**STAR Topology**

Real-world example: **online banking portals** (e.g., HDFC NetBanking) — every branch/ATM terminal connects to a central server, so the bank can monitor and secure all transactions from one point, and one faulty terminal never disrupts the others.

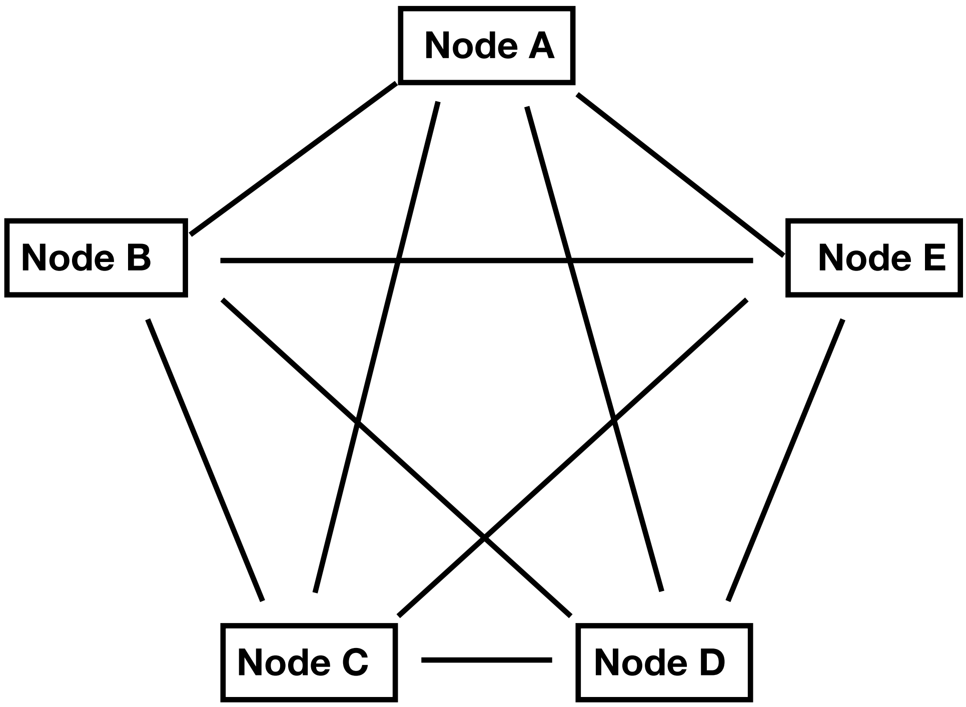
**2.**




**BUS Topology**

Real-world example: small local cable-TV / early cyber-café LAN setups — cheap to wire a single line through a row of desks, though rarely used today because one cable break takes the whole network down

**3.**




**MESH topology**

Real-world example: IPL live streaming backbone / internet backbone routers — core routers of ISPs are meshed so traffic can reroute instantly if one link goes down, keeping the stream live for millions.

## **Task 2:**

| **Topology** | **Pros** | **Cons** | **Best Use Case** |
|---|---|---|---|
| **Star** | 1. Easy to add/ remove devices<br>2. One faulty cable doesn't crash the network<br>3. Centralized management/ monitoring<br>4. Easy to troubleshoot (check the hub) | 1. Hub is a single point of failure<br>2. More cable needed than bus<br>3. Hub cost adds up<br>4. Hub performance limits network speed | Office LANs, home Wi-Fi, retail store networks.<br>Eg: **Zomato's** restaurant-side POS terminals connecting to one in-store router |
| **Bus** | 1. Very low cabling cost<br>2. Simple to set up for small networks<br>3. Good for temporary/ small setups<br>4. Uses less cable than star | 1. Entire network fails if backbone cable breaks<br>2. Hard to troubleshoot (fault location unclear)<br>3. Slows down as more devices join (collisions)<br>4. Poor security — all data passes every node | Very small, temporary, low-budget setups<br>Eg: Old cyber-café LANs (rarely used by modern apps) |

| Topology | Pros | Cons | Best Use Case |
|---|---|---|---|
| **Ring** | 1. Data flows in a predictable, orderly manner<br>2. No collisions (token-based)<br>3. Performs well under heavy, steady load<br>4. Easier to manage than a bus | 1. One broken link can disrupt the whole ring (unless dual-ring)<br>2. Adding/ removing a node disrupts traffic<br>3. Harder to troubleshoot than star<br>4. Slower than star for random-access data | Older telecom/ campus networks needing orderly, predictable data flow<br>Eg : Legacy **airline reservation systems** using token-ring style orderly transaction processing |
| **Mesh** | 1. Extremely reliable — multiple paths if one fails<br>2. High data security (dedicated links)<br>3. No traffic congestion at a central point<br>4. Great fault isolation | 1. Very expensive (lots of cabling/ hardware)<br>2. Complex to install and maintain<br>3. Redundant links waste bandwidth capacity<br>4. Hard to scale to very large networks | Critical infrastructure, backbone networks, apps needing near-zero downtime<br>Eg :<br>**Paytm's** payment-gateway server clusters, meshed so a transaction always finds a live path even if one server node fails |

## <u>Task 3:</u>

### Topology for a PUBG/Free Fire-style multiplayer game

A star topology (with the "hub" being cloud data-center servers) is the right choice for the game servers — each player's device connects directly to a central authoritative game server rather than to other players. This keeps speed high because the server can instantly validate and broadcast game state without waiting for peer-to-peer relays, improves reliability since the server enforces consistent rules and prevents cheating/desync, and keeps cost manageable because you only need to provision and scale the central servers (in regional data centers) rather than build a full mesh between every player.

## Task 4:

### Warehouse with 8 computers (Flipkart-style)

Recommended topology: Star topology, with all 8 computers connected to one central switch. This is best because it's cheap and simple to set up for a small warehouse, easy to troubleshoot (a fault light on the switch shows exactly which cable/PC is at fault), and easy to expand later — adding a 9th or 10th computer just means plugging into a free switch port, with no disruption to the rest of the order-processing network.